

Course Title	Advanced Logic for Computer Science		
Course Code	IT543	Credit Structure	3-0-0-3 (L-T-P-C)
Category	Elective	Semester	Autumn Semester (AY 25-26)
Program	M.Tech(ICT)		
Prerequisites	Discrete Mathematics		
Course Objectives/ Brief Course Description	At the end of the course, students are expected (i) To obtain basic understanding of most known modal logics and their applications (ii) To know the nature of tableaux method for decidability (iii) To understand different properties of models in modal logic (iv) To learn various systems of modal logic and how to apply them to solve problems. (v) To write a scientific report about an application of modal logic in AI, distributed systems		
Evaluation/ Grading Policy	Midsem 25% Endsem 45% Project/Presentation 20% Quizes 10%		
Course Materials/ References	Textbook: (i) Modal Logic, P Blackburn, M. de Rijke, and Yde Venema, Cambridge University Press (ii) Modal Logic, A. Chagrov, M. Zakharyashev, Oxford Science Publications (iii) Mathematical Logic for Computer Science, M. Ben-Ari, Springer.		
Detailed Course Content	Syllabus: Propositional Logic, Syntax and Semantics, Semantic tableau, Hilbert proof system, Basic properties of propositional logic, Intuitionistic logic, Kripke frames and models, Truth preserving operations, Hintikka systems, Intuitionistic frames and formulas,, Intuitionistic calculus, Basic properties of intuitionistic logic, Modal logic, Possible world semantics, Modal frames and models, Truth preserving operations, Hintikka systems, Modal frames and formulas,		

	Calculus K, Basic properties of K, A few more modal logics, From logics to classes of logics, Completeness theorems, Canonical models and filtration, Henkin construction, filtration method, Incompleteness, First Order Logic, Syntax and semantics, Hintikka systems, first order calculus
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Course Outcome:

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
x	x		x	x							x

Course Plan(Tentative)

Sl No	Topics	Lectures
1	Propositional Logic, Syntax and Semantics, Semantic tableau, Hilbert proof system, Basic properties of propositional logic,	6
2	Intuitionistic logic, Kripke frames and models, Truth preserving operations, Hintikka systems, Intuitionistic frames and formulas,, Intuitionistic calculus, Basic properties of intuitionistic logic,	10
3	Modal logic, Possible world semantics, Modal frames and models, Truth preserving operations, Hintikka systems, Modal frames and formulas, Calculus K, Basic properties of K, A few more modal logics, From logics to classes of logics,	12
4	Completeness theorems, Canonical models and filtration, Henkin construction, filtration method, Incompleteness,	5
5	First Order Logic, Syntax and semantics, Hintikka systems, first order calculus	5